



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2024-25
COURSE HANDOUT

Date: 31.07.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : CHE G558
Course Title : Chemical Process Optimization
Instructor-in-Charge : AJAYA KUMAR PANI

1. Course Description:

Introduction to Process Modelling and simulation, Fundamentals of analytical optimization. Survey of one dimensional line-search methods, and multidimensional unconstrained and constrained numerical optimization algorithms. Applications of linear programming, nonlinear programming, mixed integer linear/ nonlinear programming, and parameter estimation in chemical engineering. Feasible-path and infeasible path techniques for chemical process flowsheet optimization, Evolutionary computation in Chemical Engineering.

2. Scope and Objective of the Course:

The chemical industry has undergone significant changes during the past 25 years due to the increased cost of energy, increasingly stringent environmental regulations, and global competition in product pricing and quality. One of the most important engineering tools for addressing these issues is optimization. Optimization as the name suggests, deals with finding the optimum design parameters for maximizing the profit or minimizing the total investment on a process plant. The criteria for analysis of the economic objective function involve fulfilling a process criteria restrictions, conditions, design equations, and respecting the limits of the variables. Most problems in chemical engineering processes have many solutions, in some cases becoming endless. The optimization is related to the selection of an option that is best in a variety of efficient options but being the only one that comes closest to an economic optimum performance and operation. This course, as a subject, deals with the concepts of modeling & simulation, various methods of optimization, objective functions, constraints, theoretical & computational basis for optimization techniques, various chemical engineering examples, etc.

The objective of the course is to apply the optimization techniques in chemical engineering processes for reducing the operating cost and maximizing the profit. After completing this course the students will be able to apply the principle of optimization in solving engineering problems and understand the scope of optimization in solving Chemical Engineering problems.

3. Text Books:

TB: Thomas F. Edgar, David M. Himmelblau, "Optimization of Chemical Processes", Second Edition, McGraw Hill, New York, 1998.

4. Reference Books:

R1: Kalyanmoy Deb, "Optimization for Engineering Design: Algorithms and Examples", Second Edition, PHI Learning Private Limited, New Delhi, 2012.

5. Course Plan:

Module No. & Topic (Lecture range)	Lecture session	Ref. (TB)	Learning Outcome
1.The Nature and Organization of Optimization Problems	L1.1 Optimization definition,	Ch.1	Understanding of the procedure for solving optimization problems.
	L1.2 Scope and hierarchy of optimization, Examples,		
	L1.3 General procedure for solving optimization problem		



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(1-3)			
2.Developing Models for Optimization (4)	L2.1 Selecting functions to fit experimental data, Factorial experimental design, Degree of freedom, Examples of inequality and equality constraints in models	Ch. 2	Understanding how to develop models from experimental data
3.Formulation of Objective Functions (5)	L3.1 Economic objective functions, Time value of money in objective functions, Measures of probability	Ch. 3	Understanding the formulation of objective function for a an optimization problem
4.Basic Concepts and Types of Optimization (6-8)	L4.1 Limitations and applications of optimization, Types of optimization, Analytical methods of optimization,	Ch. 4	Understanding the basic concepts of optimization.
	L4.2. Continuity of functions, NLP problem statement, Convexity and its applications		
	L4.3. Interpretation of objective functions, Conditions for an Extremum of an unconstrained function		
5.Optimization of Unconstrained Functions: One-Dimensional Search (9-11)	L5.1 Numerical methods for optimizing a function of One Variable, Scanning and bracketing procedures	Ch. 5	Understanding the principles of one dimensional search.
	L5.2. Newton and Quasi-Newton methods of unidimensional Search		
	L5.3. Polynomial approximation methods, Evaluation of unidimensional search methods		
6.Unconstrained Multivariable Optimization (12-13)	L6.1. Methods using function values only, Methods that use first derivatives	Ch. 6	Understanding the principles of unconstrained multivariable search.
	L6.2. Newton's method, Quasi-Newton Methods		
7.Linear Programming (LP) and Applications (14-16)	L7.1. Geometry of linear programs, Basic linear programming	Ch. 7	Understanding linear programming problems.
	L7.2. Simplex algorithm, Barrier methods, Sensitivity analysis		
	L7.3. Linear mixed integer programs, LP software, Network flow and assignment problems		
8.Nonlinear Programming Constraints with (17-20)	L8.1. Direct Substitution, First-order necessary conditions for a local extremum, Quadratic programming,	Ch. 8	Understanding non-linear programming problems.
	L8.2. Penalty, Barrier, and Augmented Lagrangian Methods, Successive Linear Programming,		
	L8.3 Successive quadratic programming, The generalized reduced gradient method,		
	L 8.4 Relative advantages and disadvantages of NLP Methods, Available NLP software		
9.Mixed-Integer Programming (21-24)	L9.1 Problem Formulation, Branch-and-bound methods using LP relaxations	Ch.9	Understanding the principles of mixed-integer non-linear programming.
	L9.2 Solving MINLP problems using branch-and-bound methods		
	L9.3 Solving MINLPs using outer approximation		
	L9.4 Other decomposition approaches for MINLP, Disjunctive programming		
10.Global Optimization for Problems with Continuous and Discrete Variables (25-29)	L10.1 Methods for global optimization	Ch.10	Understanding global optimization problems.
	L10.2 Smoothing optimization problems		
	L10.3 Branch-and-bound methods		
	L10.4 Multistart methods		
	L10.5 Heuristic search methods		



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11.Applications of Optimization (30-35)	L11.1 Heat transfer and energy conservation	Ch.11	Understanding the applications of optimization in Chemical Engineering problems.
	L11.2 Separation processes	Ch.12	
	L11.3 Fluid flow systems	Ch.13	
	L11.4 Chemical reactor design and operation	Ch.14	
	L11.5 Optimization in large-scale plant design and operation	Ch.15	
	L11.6 Integrated planning, scheduling and control in the process industries	Ch.16	
12.Evolutionary computation in Chemical Engineering (36-40)	L12.1 Genetic Algorithm		Understanding application of evolutionary computation in Chemical Engineering Problems.
	L12.2 Genetic Algorithm		
	L12.3 Differential Evolution		
	L12.4 Differential Evolution		
	L12.5 Particle Swarm Optimization		

(N.B. The order of the lecture modules may be altered based on the requirements.)

6. Evaluation Scheme:

Component	Duration (Minutes)	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90	30		CB
Project and Seminar	-	20	-	-
Declared class Tests (Best 2 of 3)	-	10		
Comprehensive Examination	180	40		CB and/or OB

7. Chamber/Online Consultation Hour: Tuesday 4-5 PM.

8. Notices: All notices concerning this course will be displayed in Google Classroom for this course.

9. Make-up Policy:

If the student is unable to appear for the Regular Test/Examination due to genuine exigencies, the student must refer to the procedure for applying for Make-up Test/Examination. No make-up will be granted for the tutorials and surprise class tests.

10. Note (if any):

It shall be the responsibility of the individual student to be regular in attending the lectures and to be updated oneself on the subjects covered by solving relevant questions and examples.

Instructor-in-charge
Course No. CHE G558